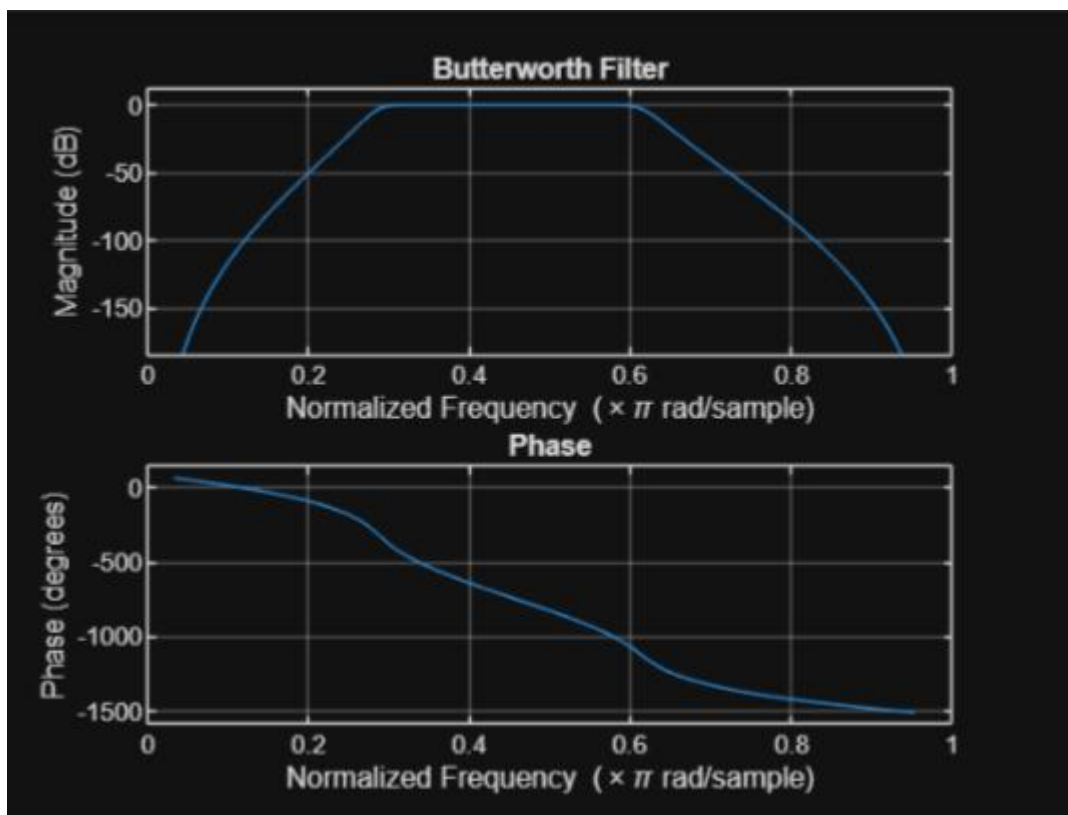
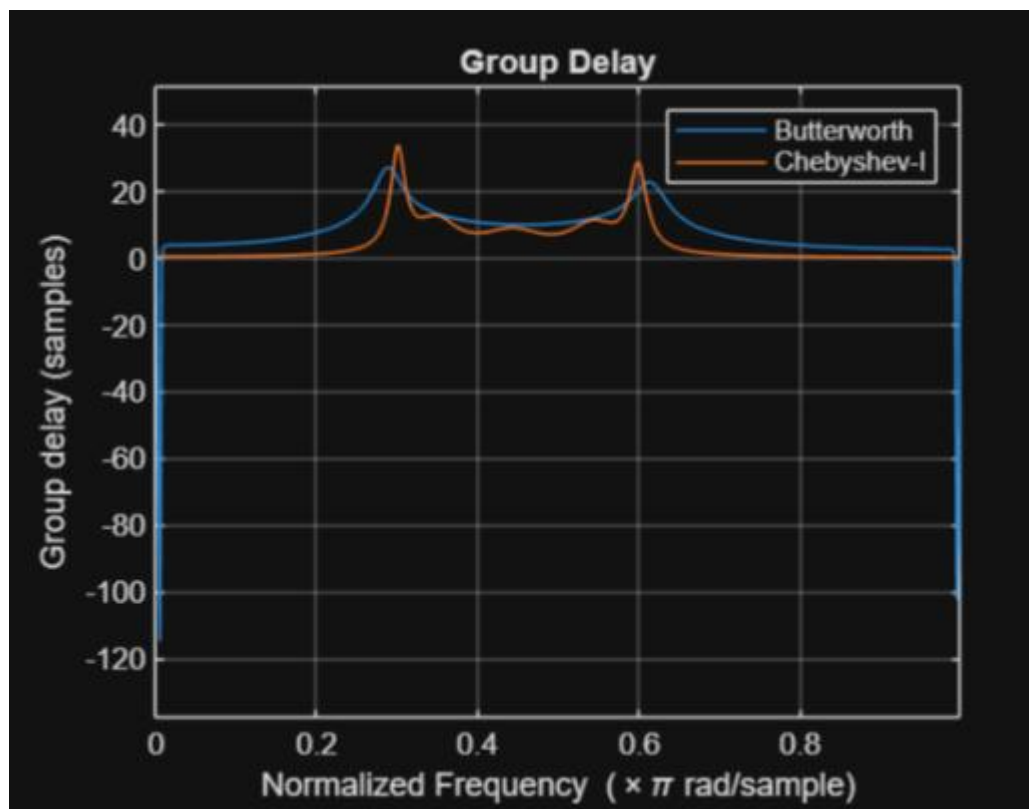
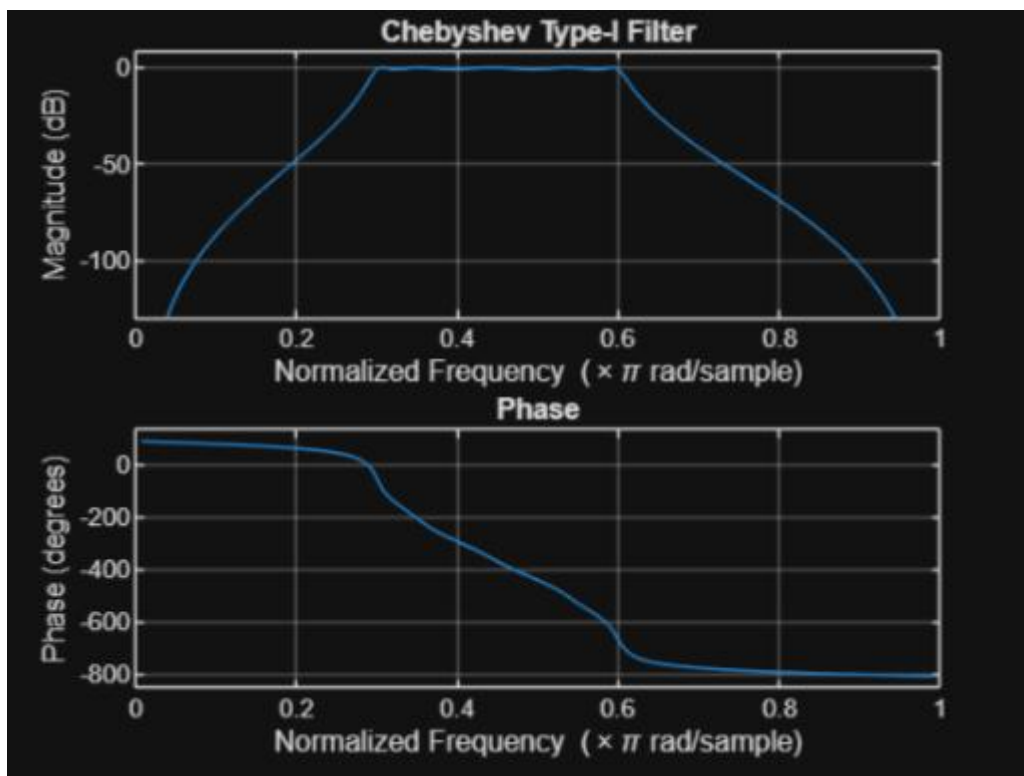


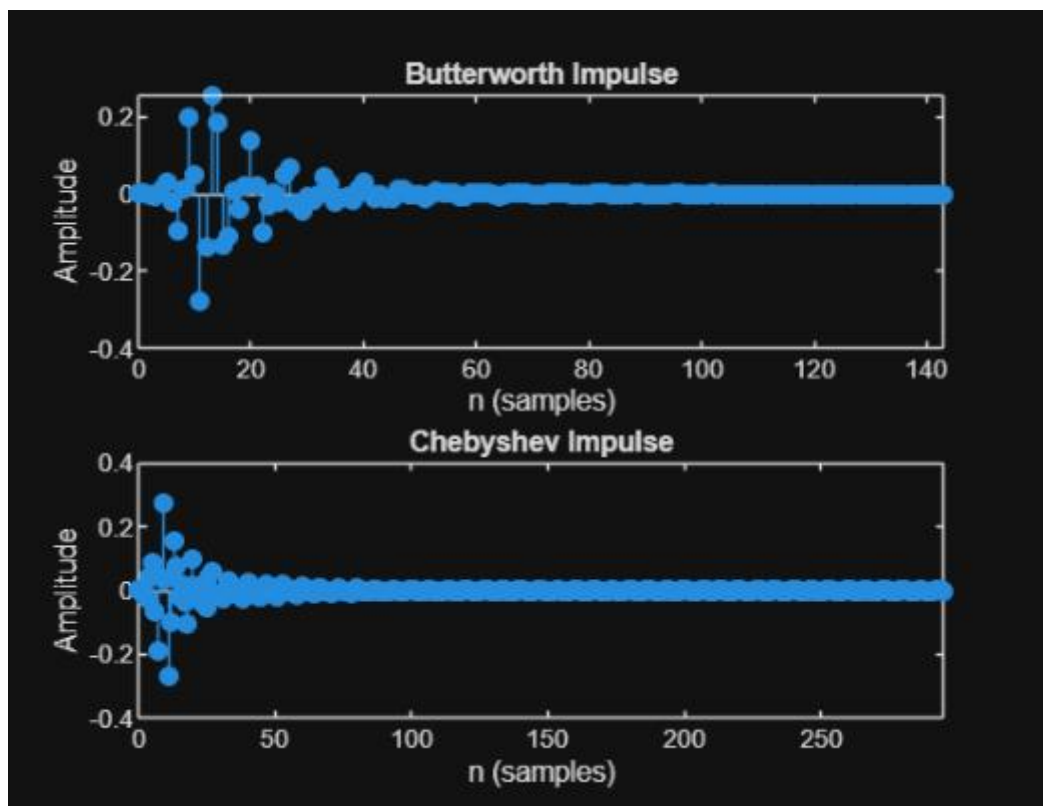
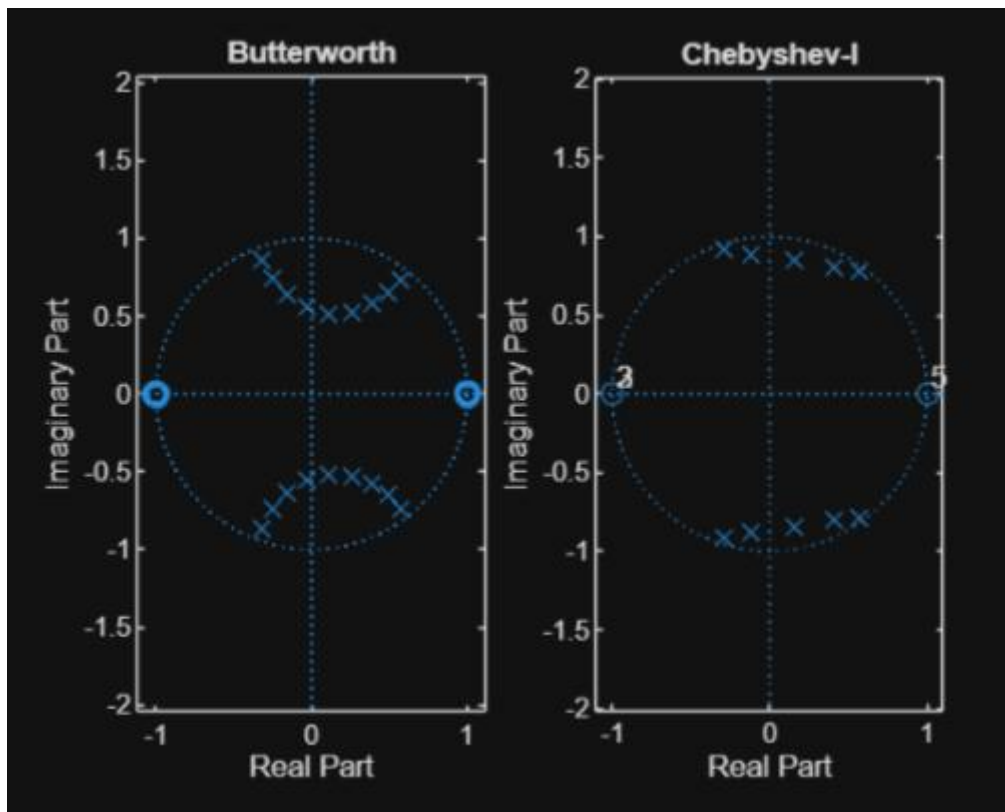
Exp 3 Performance Analysis of IIR Filters

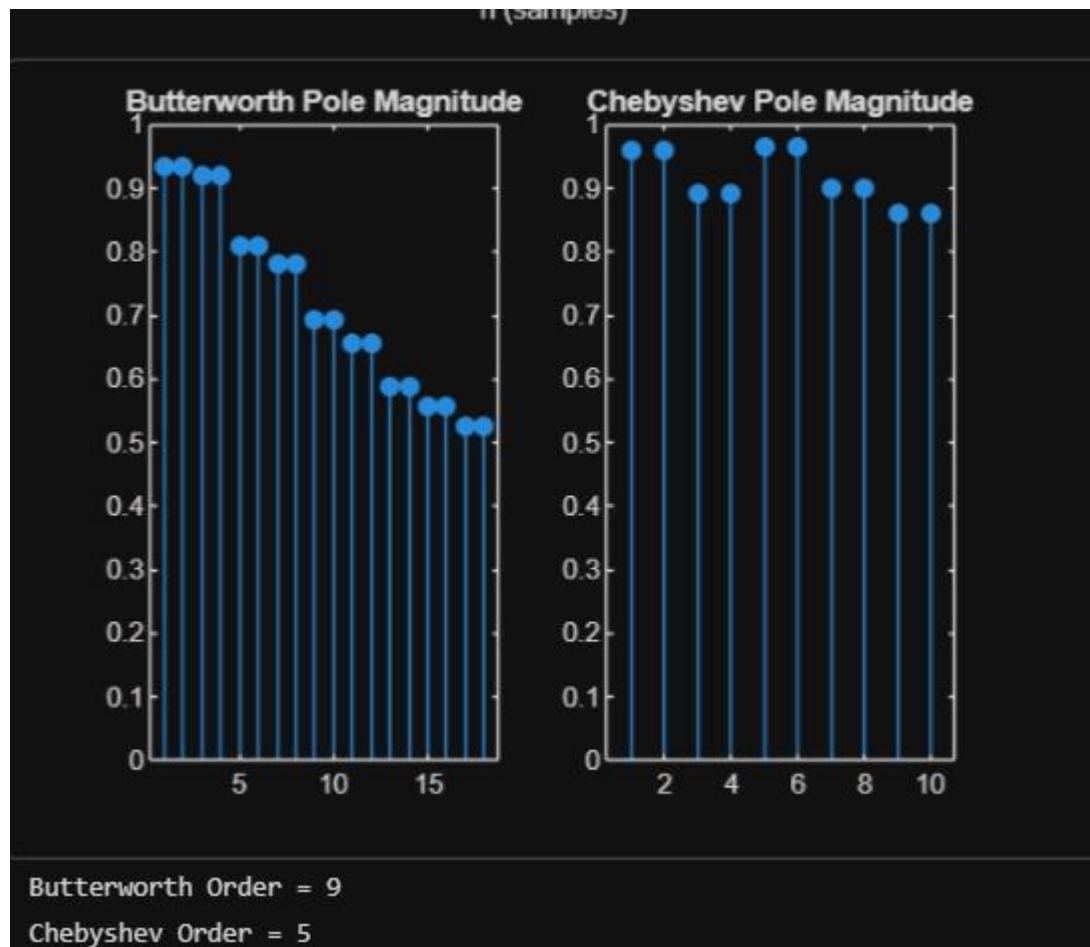
3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

```
1  clc; clear; close all;
2  % Specifications
3  Fs=2000; Rp=1; Rs=40;
4  Wp=[300 600]/(Fs/2);
5  Ws=[200 700]/(Fs/2);
6  % Butterworth Filter
7  [n1,w1]=buttord(Wp,Ws,Rp,Rs);
8  [b1,a1]=butter(n1,w1,'bandpass');
9  % Chebyshev Type-I Filter
10 [n2,w2]=cheb1ord(Wp,Ws,Rp,Rs);
11 [b2,a2]=cheby1(n2,Rp,w2,'bandpass');
12 % Magnitude Response
13 figure;
14 freqz(b1,a1); title('Butterworth Filter');
15 figure;
16 freqz(b2,a2); title('Chebyshev Type-I Filter');
17 % Group Delay
18 figure;
19 grpdelay(b1,a1); hold on;
20 grpdelay(b2,a2);
21 legend('Butterworth','Chebyshev-I');
22 % Pole-Zero Plot
23 figure;
24 subplot(121),zplane(b1,a1),title('Butterworth')
25 subplot(122),zplane(b2,a2),title('Chebyshev-I')
26 % Impulse Response
27 figure;
28 subplot(211),impz(b1,a1),title('Butterworth Impulse')
29 subplot(212),impz(b2,a2),title('Chebyshev Impulse')
30 % Pole Magnitude
31 figure;
32 subplot(121),stem(abs(roots(a1)),'filled')
33 title('Butterworth Pole Magnitude')
34 subplot(122),stem(abs(roots(a2)),'filled')
35 title('Chebyshev Pole Magnitude')
36 disp(['Butterworth Order = ',num2str(n1)])
37 disp(['Chebyshev Order = ',num2str(n2)])
```









Results:

1. Butterworth and Chebyshev Type-I IIR Band-Pass Filters were successfully designed and implemented in MATLAB.
2. The minimum filter orders and cutoff frequencies were obtained for both filters.
3. The frequency responses were analyzed and compared; the Chebyshev filter provided a sharper transition, while the Butterworth filter showed a smoother response.
4. Pole-zero analysis confirmed that both filters are stable and satisfy the given design specifications.